

Physical Analog Pandrosion for Continuous-Time Root Extraction

A corrected circuit-physics version of the legacy analog paper

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Abstract

This paper rebuilds the legacy note *Analog Pandrosion* as a physics-facing research article. The original note proposed a lookup-free and clockless analog architecture for computing p -th roots from the monomial Pandrosion map

$$h_{p,y}(s) = 1 - \frac{y - 1}{y(1 + s + \dots + s^{p-1})}.$$

The main correction is one of scope. Analog root extraction is not new: log-domain and computational-unit devices already implement products, quotients, powers, and roots. The contribution claimed here is narrower and testable: Pandrosion gives a repeated, derivative-free, positive-domain h -cell whose residual multiplier is controlled by the chart-normalized target y , and whose physical error can be bounded by the same normalization and atlas principles developed in Papers 000–023.

We formulate a dimensionally consistent signal model, a positive-current log/translinear implementation, and a voltage-mode macrocell implementation. The chart selector from the monomial and atlas papers keeps $y \geq 1$ and $|\log y|$ small, avoiding sign changes in the numerator and keeping the denominator $yS_p(s)$ away from zero. For a normalized target satisfying $1 \leq y \leq q^{p/(2K)}$, the fixed-point multiplier is bounded by

$$\Lambda_{p,q,K} = 1 - \frac{p}{S_p(q^{1/(2K)})}.$$

This gives a direct small-signal stability estimate for deterministic component offsets, thermal noise, multiplier gain error, divider error, and finite settling. A repeated h -cell pipeline is therefore not presented as an operation-minimal replacement for digital Newton or for existing analog computational units. It is presented as a modular physical realization of the Pandrosion geometry: a continuous-time normalized root cell whose accuracy is set by chart normalization, component nonidealities, and settling bandwidth.

One-sentence summary. The improved analog claim is not “first analog root circuit”; it is “a positive-domain, chart-normalized, repeated Pandrosion h -cell with a closed residual multiplier and explicit physical error budget.”

Status ledger. The fixed-point, chart-normalization, and multiplier bounds are mathematical consequences of Papers 000–002 and 009–010. The circuit models below are architectural physics models, not a fabricated silicon result. SPICE, layout parasitics, temperature drift, and measured bandwidth remain required before any hardware performance claim should be made.

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1 Why the legacy analog paper needs a rebuild

The legacy April 2026 note made a strong claim: a Pandrosion pipeline computes p -th roots with additions, multiplications, and division, without derivatives, lookup tables, stored coefficients, or a clock. That core idea is worth preserving, but three statements need to be sharpened.

1. Analog root extraction already exists. Classical log-domain, translinear, and real-time computational-unit circuits can compute products, quotients, powers, and roots. The AD538, for example, is explicitly marketed as a real-time analog computational unit with resistor-programmable powers and roots. Therefore the novelty cannot be “analog root extraction exists”.
2. Operation count is not a sufficient physical metric. A single Pandrosion h -cell for $p = 3$ uses a polynomial ladder, a multiplier, a divider, and a subtractor. A useful pipeline repeats that cell. The relevant metrics are error budget, bandwidth, stability margin, dynamic range, offset, temperature drift, and calibration complexity.
3. “Clockless” means continuous-time settling, not infinite-speed computation. A combinational analog chain has propagation delay, finite bandwidth, parasitic poles, slew limits, and output settling error. If the result is consumed by a digital system, a sample-and-hold or ADC introduces a clock at the interface.

The improved claim is consequently narrower. Pandrosion supplies a repeated cell with one algebraic shape, no derivative block, no table lookup, and a closed residual multiplier. The physical question is whether this repeated cell can be made accurate and stable enough, in the target bandwidth and power range, to be useful.

2 Inputs imported from Papers 000–023

The present paper reads the current May 2026 series as the mathematical backbone for a corrected physics paper. The following table records what is used from each article.

Paper	Role in the analog rebuild
000	Monomial Pandrosion map, fixed point, exact multiplier, Thales scaling, reciprocal chart, scale-palette theorem.
001	Chart-scaled finite-slope solver ladder and the root-coordinate normalization $u^p = y$.
002	Divided-difference interpretation and fixed-anchor residual law.
003	Chart optimization and local curvature cancellation.
004	Higher-order jet cancellation and the inverse-chart principle.
005	Tensorial multidimensional reading of finite-slope blocks.
006	Equal-value pole geometry, relevant to avoiding divider singularities.
007	Variational chart geometry and distributed flattening over a domain.
008	Finite-slope descent and Lyapunov viewpoint for accepted corrections.
009	Dynamic atlas language: chart scores, homothety, inversion, transition rules.
010	Stable numerical atlases and projective conditioning.
011	Probe paths and stability metrics.
012	Low-action/geodesic interpretation of stable probes.
013	Conditional geometric descent statements.
014	All-order tensor inverse jets.
015	Uniform finite atlas and anchor-palette logic.
016	Analytic convergence under Lojasiewicz-type descent.
017	Finite-probe reconstruction as a derivative-free physical measurement model.
018	Equal-value pole monodromy and pole-collision caution.
019	Discrete probe convergence, relevant to finite sampled hardware probes.
020	Finite-slope polynomial-system framework and honest conditional status.
021	Average-polynomial formulation and failure-probability amplification style.
022	Universal atlas viewpoint and fixed-cell geometry.
023	Hypercube design, random-matrix stabilization, and explicit proof ledger discipline.

The analog paper uses only the monomial positive-real part as a proposed hardware primitive. The later atlas and finite-probe papers enter as design discipline: normalize first, avoid equal-value poles, keep charts well conditioned, and separate proved statements from experiments.

3 Mathematical core of the root cell

Let $p \geq 2$ and $y > 0$. Define

$$S_p(s) = 1 + s + \cdots + s^{p-1}, \quad h_{p,y}(s) = 1 - \frac{y-1}{yS_p(s)}.$$

For $y \geq 1$, the positive fixed point is

$$s_* = y^{-1/p}, \quad u_* = ys_*^{p-1} = y^{1/p}.$$

Thus s is the internal reciprocal-root coordinate and u is the visible root-coordinate readout.

Proposition 3.1 (Fixed point and readout). *For $y > 0$, $s_* = y^{-1/p}$ is a fixed point of $h_{p,y}$, and*

$$ys_*^{p-1} = y^{1/p}.$$

Proof. At $s_* = y^{-1/p}$, one has

$$yS_p(s_*) = \frac{y(1 - s_*^p)}{1 - s_*} = \frac{y - 1}{1 - s_*}$$

when $s_* \neq 1$, with the case $s_* = 1$ obtained by continuity. Hence

$$1 - \frac{y-1}{yS_p(s_*)} = s_*.$$

The readout identity follows from $s_*^p = 1/y$. □

Proposition 3.2 (Exact local multiplier). *Put $\alpha = y^{1/p}$. At the positive fixed point,*

$$\lambda_{p,y} = h'_{p,y}(s_*) = 1 - \frac{p}{1 + \alpha + \dots + \alpha^{p-1}}.$$

In particular, $\lambda_{p,1} = 0$, and $\lambda_{p,y}$ grows as y moves away from 1 on the branch $y \geq 1$.

This multiplier is the main reason an analog implementation should include a chart/scale selector before the h -pipeline. A raw circuit forced to handle all magnitudes in one affine chart has a poor residual multiplier and a poor physical dynamic range.

4 Dimensional normalization and chart selection

Physical voltages and currents have units. The map $h_{p,y}$ is dimensionless. Therefore a root extractor must first convert the input signal to a dimensionless normalized target.

Let $X(t) > 0$ be the physical input quantity and let $B > 0$ be a physical root-scale, with units chosen so that B^p has the units of X . There are two positive charts:

$$y_+(B) = \frac{X}{B^p}, \quad y_\infty(B) = \frac{B^p}{X}.$$

The direct chart returns $Z = Bu$. The reciprocal chart returns $Z = B/u$. Both are physically valid because

$$u^p = y_+(B) \Rightarrow (Bu)^p = X, \quad u^p = y_\infty(B) \Rightarrow (B/u)^p = X.$$

Choose a root-scale palette

$$\mathcal{B}_{q,K} = \{q^{m+j/K} B_0 : m \in \mathbb{Z}, j = 0, \dots, K-1\}.$$

The selector chooses a chart and scale such that $y \geq 1$ and $\log y$ is minimal. The monomial scale-palette theorem gives

$$1 \leq y \leq q^{p/(2K)}, \quad 1 \leq y^{1/p} \leq q^{1/(2K)}.$$

Theorem 4.1 (Normalized physical multiplier bound). *Assume the chart selector gives $1 \leq y \leq q^{p/(2K)}$. Then the local residual multiplier of the analog h -cell satisfies*

$$0 \leq \lambda_{p,y} \leq \Lambda_{p,q,K} := 1 - \frac{p}{1 + q^{1/(2K)} + \dots + q^{(p-1)/(2K)}}.$$

Proof. The multiplier formula depends monotonically on $\alpha = y^{1/p} \geq 1$. The selector gives $\alpha \leq q^{1/(2K)}$. Substitution gives the bound. □

This theorem is the hardware version of the chart-scaling message: the physical cell should not be asked to compute roots over decades of input range without first moving the problem into a bounded root-coordinate annulus.

5 The h -cell as a physical block

For a fixed normalized target $y \geq 1$, an h -cell maps an input s to

$$s^+ = 1 - \frac{y-1}{yS_p(s)}.$$

The cell contains four conceptual subblocks:

$$s \mapsto S_p(s), \quad (y, S_p) \mapsto yS_p, \quad (y-1, yS_p) \mapsto \frac{y-1}{yS_p}, \quad r \mapsto 1-r.$$

The numerator $y-1$ is shared across repeated stages for a fixed input sample or slowly varying input trajectory. It should not be counted once per stage in a physical layout unless each stage has its own independent input conditioner.

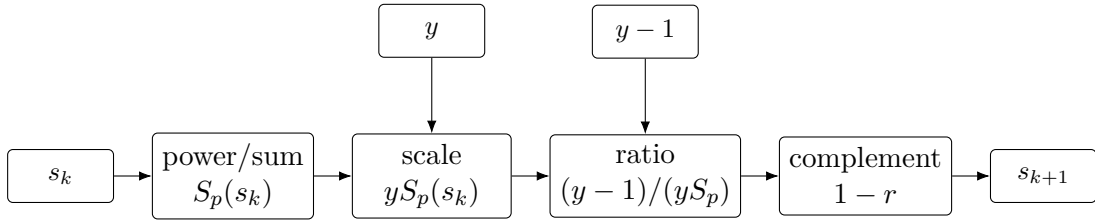


Figure 1: One normalized Pandrosion h -cell. Repeated pipelines copy this cell while sharing the normalized input y and numerator $y-1$.

5.1 Current-mode log/translinear realization

The cleanest physics model uses positive currents. Encode each positive dimensionless variable a by a current $I_a = I_0 a$, or in a log-domain variant by a voltage $V_a = U_T \log a$ where $U_T = kT/q_e$. Sums are natural in current mode:

$$I_{S_p} = I_0 + I_s + I_{s^2} + \cdots + I_{s^{p-1}}.$$

Products and quotients can be formed by translinear loops or log/antilog blocks:

$$I_{ab} = \frac{I_a I_b}{I_0}, \quad I_{a/b} = \frac{I_a I_0}{I_b}.$$

The Pandrosion denominator is strictly positive:

$$I_{yS_p} \geq I_y > 0,$$

so the divider is not exposed to the equal-value pole inside the intended positive chart.

Remark 5.1 (Temperature). *Log-domain circuits inherit explicit thermal-voltage dependence. A raw implementation is therefore temperature sensitive. A credible design must use matched devices, ratioed currents, bias tracking, or calibration. The Pandrosion geometry removes derivative and lookup blocks; it does not remove semiconductor physics.*

5.2 Voltage-mode macrocell realization

For a board-level demonstrator, the same block can be built from analog multipliers, op-amp summers, and a divider. Devices such as AD633 and AD835 are representative multiplier macrocells; AD538 is a representative analog computational unit that already exposes powers and roots. This matters for positioning: a Pandrosion board should be compared against existing analog computational units, not only against digital Newton.

For $p = 3$, a shared-input h -pipeline uses per cell:

$$s^2, \quad 1 + s + s^2, \quad y(1 + s + s^2), \quad \frac{y - 1}{y(1 + s + s^2)}, \quad 1 - \frac{y - 1}{y(1 + s + s^2)}.$$

The readout uses $u = ys_N^2$, followed by $Z = Bu$ in the direct chart or $Z = B/u$ in the reciprocal chart.

6 Noise, offset, and settling model

Let the physical cell be modeled as

$$\tilde{h}(s) = h_{p,y}(s) + \beta(s, y) + \xi(s, y),$$

where β is a deterministic bias term caused by gain error, offset, nonlinearity, and finite settling, while ξ is zero-mean random noise. Near the fixed point,

$$e_{k+1} = \lambda_{p,y} e_k + \beta_* + \xi_k + \mathcal{O}(e_k^2), \quad e_k = s_k - s_*.$$

Assumption 6.1 (Lumped cell error). *In the operating annulus selected by the chart layer, suppose*

$$|\beta(s, y)| \leq \beta_{\text{cell}}, \quad \text{Var}(\xi(s, y)) \leq \sigma_{\text{cell}}^2, \quad |h'_{p,y}(s)| \leq \Lambda < 1.$$

Theorem 6.2 (Pipeline error budget). *Under the lumped cell assumption, the N -cell pipeline satisfies, to first order in the cell errors,*

$$|\mathbb{E}e_N| \leq \Lambda^N |e_0| + \frac{1 - \Lambda^N}{1 - \Lambda} \beta_{\text{cell}},$$

and, for independent stage noise,

$$\text{Var}(e_N) \leq \frac{1 - \Lambda^{2N}}{1 - \Lambda^2} \sigma_{\text{cell}}^2.$$

Proof. Iterate the affine stochastic recurrence

$$e_{k+1} = \Lambda e_k + \beta_k + \xi_k$$

with $|\beta_k| \leq \beta_{\text{cell}}$. The expectation is bounded by a geometric series. Independence and zero mean give the variance series. $\square$

Corollary 6.3 (Why normalization is a physical requirement). *If the selector makes $\Lambda = \Lambda_{p,q,K} \ll 1$, both the deterministic initial residual and the accumulated physical errors are suppressed by short geometric sums. If y is not normalized and $\Lambda \approx 1$, offsets, thermal noise, and settling error accumulate almost linearly with pipeline depth.*

The legacy paper emphasized mathematical bias. The corrected physical version separates three different floors:

$$\text{math residual} + \text{deterministic circuit bias} + \text{random noise}.$$

Post-filtering can reduce random noise but cannot remove a deterministic mathematical residual or a systematic circuit offset.

7 Deterministic bias of the normalized h -pipeline

The following table gives exact floating-point evaluations of the ideal mathematical h -pipeline for $y = 2$, initialized by the physical preconditioning stage $s_0 = h_{p,2}(1)$. The visible output is $u_N = y s_N^{p-1}$, and the error is $|u_N - 2^{1/p}|$.

p	stages 1	stages 2	stages 3	stages 4
2	$8.58 \cdot 10^{-2}$	$1.44 \cdot 10^{-2}$	$2.45 \cdot 10^{-3}$	$4.21 \cdot 10^{-4}$
3	$1.29 \cdot 10^{-1}$	$2.71 \cdot 10^{-2}$	$5.91 \cdot 10^{-3}$	$1.30 \cdot 10^{-3}$
5	$1.64 \cdot 10^{-1}$	$3.93 \cdot 10^{-2}$	$9.92 \cdot 10^{-3}$	$2.53 \cdot 10^{-3}$

These numbers should not be read as a hardware benchmark. They are a deterministic residual table for the ideal map. A real circuit adds the offset/noise/settling budget of the previous section. The table is useful because it tells the circuit designer how much mathematical residual remains before component physics is added.

8 Comparison with Newton and analog computational units

Newton's method for $u^p = y$ uses

$$u^+ = u - \frac{u^p - y}{pu^{p-1}}.$$

It has excellent mathematical convergence when supplied with a good seed, but the hardware block contains explicit p -dependent gain and a derivative denominator. In a digital processor this is not a problem; in a fixed analog cell it changes the calibration surface.

The Pandrosion h -cell instead places the p -dependence in the polynomial ladder S_p . This is not automatically cheaper. It is a different physical trade:

Newton: fewer iterations, derivative/gain-specific cell, seed sensitive;
 Pandrosion: repeated identical cells, derivative-free, chart-normalized.

Existing analog computational units provide another comparison class. An AD538-style part can directly implement powers and roots through a calibrated log-domain computation. A Pandrosion implementation must therefore justify itself through modularity, absence of a stored table/seed, positive-domain stability, or application-specific integration into a larger analog signal path. It should not claim to beat a mature computational unit without measured data.

9 Applications where the architecture is plausible

9.1 Sensor front-ends

Low-power sensor front-ends often need monotone linearization, normalization, or square-root-like compression before digitization. A normalized Pandrosion block could be useful when modest precision is enough and the root operation is embedded in a continuous signal path.

9.2 Radar and sonar magnitude channels

Magnitude extraction involves operations such as

$$|z| = \sqrt{I^2 + Q^2}.$$

For $p = 2$, the polynomial ladder is shortest and the positive-domain implementation is most attractive. The actual performance question is whether the multiplier/divider bandwidth and settling error beat a conventional rectifier/RMS/computational-unit solution for the target frequency band.

9.3 Neuromorphic and analog ML blocks

Analog neuromorphic circuits often tolerate low to moderate precision but care about locality, bandwidth, and power. Power-law activations, normalization terms, and L_p -norm approximations are plausible use cases. The main advantage would be keeping the operation inside the analog substrate rather than converting to a digital processor for a single nonlinear function.

10 Experimental program

A defensible physics paper needs experiments in the following order.

1. **Behavioral SPICE.** Implement ideal controlled-source h -cells with finite bandwidth, offset, and noise. Verify the multiplier-bound error budget.
2. **Macrocell board.** Build $p = 2$ and $p = 3$ boards using commercial multipliers/dividers or log-ratio blocks. Measure static transfer, settling time, temperature drift, and output noise.
3. **Chart selector.** Test direct and reciprocal normalization over multiple decades of X . Verify that the root-coordinate output stays in the predicted annulus.
4. **Comparison.** Compare against a direct computational-unit root circuit, a Newton-like analog macrocell, and a digital microcontroller baseline at equal bandwidth and power.
5. **ASIC feasibility.** Replace board-level macrocells by matched translinear/log-domain cells and estimate area, power, calibration overhead, and thermal sensitivity.

11 Falsifiable claims

The improved analog program should be judged by the following falsifiable claims.

1. With chart normalization, the measured small-signal residual factor of the h -pipeline tracks the theoretical bound $\Lambda_{p,q,K}$ within the measured component error budget.
2. Repeating identical h -cells gives predictable deterministic residual reduction until circuit offset or noise dominates.
3. The positive-domain denominator $yS_p(s)$ prevents internal divider singularities over the selected operating annulus.
4. For at least one target application, the complete analog block has a power-latency-precision point that is competitive with direct analog computational units or digitize-and-compute baselines.

The first three claims are near-term circuit-physics checks. The fourth is an application-level engineering claim and should not be asserted without measurements.

12 Conclusion

The legacy analog Pandrosion note contained a useful idea but overstated its hardware novelty. This rebuilt version gives the sharper claim. Pandrosion does not invent analog root extraction. It supplies a derivative-free, positive-domain, chart-normalized h -cell whose residual multiplier is known exactly and whose physical errors admit a simple pipeline budget. The right next step is not another operation-count table; it is a measured $p = 2$ or $p = 3$ demonstrator with explicit bandwidth, offset, noise, drift, and comparison baselines.

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